DO NOW — RECALL

Paging, segmentation and virtual memory

Name: _____ Class: _____ Date: _____

1. What is RAM used for?

2. What are remote virtual storage files?

3. Why might a process have to wait on memory I/O?

4. A 4-row, 3-column zero-indexed array: give its final valid index pair and number of values.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Active program and data.

2. Model answer

Files hosted on network-accessible servers.

3. Model answer

Required page/data is not resident and must be loaded.

4. Model answer

Index (3,2), twelve values.

Vocabulary — know these terms

page — Virtual fixed-size memory block

frame — Physical fixed-size RAM block

page fault — Page absent from RAM

segmentation — Logical variable-length regions

NEW LEARNING

Read, annotate and explain

Understand

Pages and frames have equal fixed sizes; the page table maps virtual pages to physical frames.

Trace

Divide the virtual address by page size for the page number; use remainder as the offset.

Explain

A page fault requires a non-resident page to be loaded. Variable-sized segmentation has different fragmentation characteristics.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Translate virtual address 520 with page size 128 and page 4 mapped to frame 9.

Predict first

520 div 128=4 remainder 8; frame 9 base=1152, plus 8 gives 1160.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define page, frame, page table and page fault. [4]

Question 2

Page size 256, VA 770, page 3 mapped to frame 5. Find physical address. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare paging and segmentation and explain thrashing. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Paging changes the offset within the page when it changes frame.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.